



INNOVATIONS IN FINANCE

LECTURE 1 : PAST, PRESENT & FUTURE

Make an impact



INTRODUCTION



Make an impact

WELCOME TO « INNOVATIONS IN FINANCE »



- ◆ Patrice Bouché, Financial Specialist for almost 30 years.
 - Degree in Actuarial Sciences and Risk Management
 - Dealing Room professional
 - ◆ Sales and structuration for french banks, working on corporate and institutionnal desks
 - Chief Investment Officer (Family Office in Geneva)
 - Structuration of a private bank in Mauritius
 - ◆ ALM, Advisory, Fx Options Trading, Crypto Fund
 - Founder of an IT company specialised in Finance, in 2017.
 - Founder of KowiKan, Financial company whose aim is to help people manage money and investments.



OUR GOAL : UNDERSTAND THE FUTURE OF FINANCE

How do you manage your money ?



HOW THIS COURSE RUNS

Eight sessions, one capstone with corporate partners — and a clear rule on AI

EIGHT SESSIONS OF TWO HOURS

- 1 **Past, Present & Future**
- 2 Payments, Lending & Neobanks
- 3 Blockchain, Tokenization & Web3
- 4 Data in Finance
- 5 AI in Finance
- 6 APIs & Embedded Finance
- 7 Regulation, Ethics & ESG
- 8 **Coding with AI — you will build something**

Plus a capstone project run with corporate partners.

USING AI IN THIS COURSE

Session 8 teaches you to code with AI. Banning it would be absurd.

Encouraged

to explore, draft, code, rephrase, and to argue with.

Declared

say where you used it. Undeclared use is the problem, not use.

Yours

you own every number and every claim you hand in. An unchecked hallucination is a mistake, not an excuse.

Except

no AI in invigilated assessments.





WHAT YOU WILL BE ABLE TO DO

Six outcomes — the last two are the ones that get assessed on the capstone

KNOW & UNDERSTAND

- LO1** Understand how fintechs and financial innovations address structural inefficiencies in traditional finance.
- LO2** Know the main emerging technologies — blockchain, AI, data, APIs, tokenization — and how data feeds AI systems.

BE ABLE TO DO

- LO3** Analyse and evaluate fintech business models: value proposition, monetization, regulatory exposure.
- LO4** Apply emerging technologies to a real financial pain point.
- LO5** **Design and prototype, in a team, a solution to a financial problem set by a corporate partner — and defend it.**
- LO6** **Assess critically the regulatory, ethical and sustainability implications of a financial innovation.**

Curiosity, creativity and critical thinking are rewarded. So is telling me I am wrong, with evidence.





PAST, PRESENT & FUTURE

When do you think the first financial innovation happened ?

- A. 20 years ago with Blockchain
- B. 70 years ago with credit cards
- C. 5,000 years ago in Mesopotamia



SESSION 1 OBJECTIVES

- ◆ Explore the historical evolution of financial innovation
- ◆ Identify inefficiencies in traditional finance systems
- ◆ Understand the drivers of fintech innovation
- ◆ Discover the way we can understand financial innovations
- ◆ Start thinking like a financial innovator



HISTORICAL PERSPECTIVE

HOW FINANCE EVOLVES

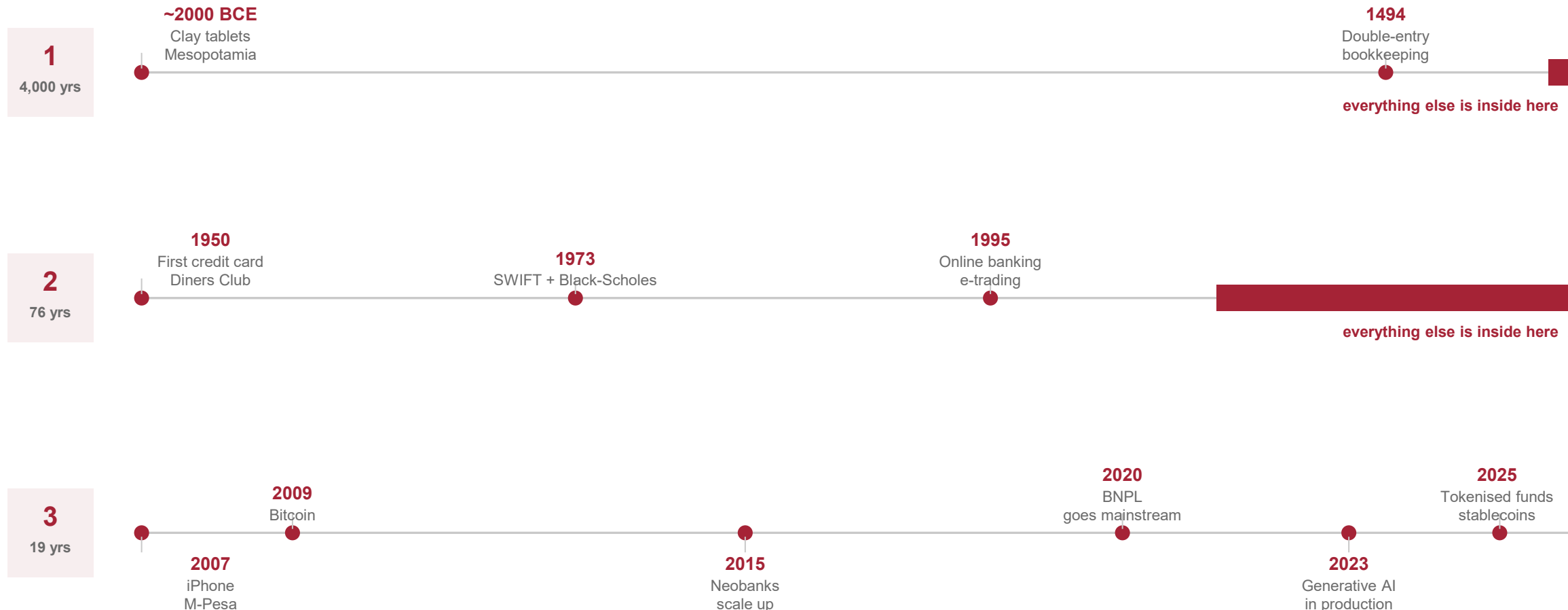


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5,000 YEARS IN ONE LOOK

Each line zooms into the red block above it — and covers about 50x less time



Same history. Real time scale. That is what acceleration looks like.



EARLY FINTECH: CLAY TABLETS AND BANKING



- ◆ Around 2000 BCE, in ancient Mesopotamia, the first forms of banking emerged in temples and palaces.
 - Clay tablets were used as promissory notes and contracts, recording deposits, loans, and interest payments.
 - Temples acted as financial institutions, facilitating lending and secure storage for merchants and farmers.
- ◆ This system introduced foundational financial concepts like credit, interest, and record-keeping — early pillars of modern finance.



THE STORY YOU WILL HEAR EVERYWHERE

It is also false — and how it became false is the more useful lesson

THE LEGEND

Nathan Rothschild keeps a private network of carrier pigeons.

He learns the outcome of the battle before anyone in London.

He trades on British government bonds ahead of the news and makes a fortune.

Told as the ancestor of high-frequency trading.

THE RECORD

- The battle is Waterloo, 1815 — not Austerlitz, 1805.
- Nathan was not on the continent on 18 June 1815.
- No evidence of any pigeon network, or of any special arrangement for that news.
- The Rothschilds did profit from the war — by financing it, not by front-running the result.
- **The story first appears in an antisemitic pamphlet by Georges Dairnvaell, 1846.**

TWO THINGS TO TAKE AWAY

1. The information edge is real. It runs from the telegraph to co-located servers. That part of the story is true — just not this story.

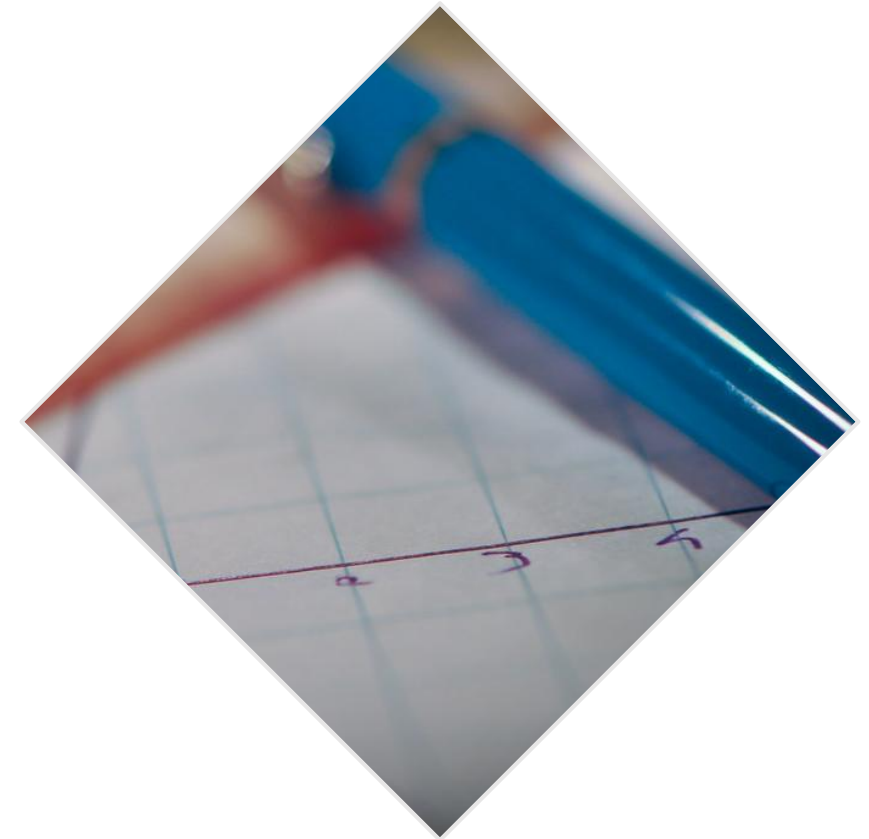
2. Finance manufactures its own myths, and they serve someone. Check the source. Including in the case studies you are handed.

Source: The Rothschild Archive, "Nathan and Waterloo".



THE BIRTH OF OPTION PRICING – BLACK & SCHOLES (1973)

- ◆ In 1973, Fischer Black and Myron Scholes published a groundbreaking model to price options, later extended by Robert Merton.
 - Before this, options were traded, but pricing was inconsistent, often based on intuition rather than rigorous methods.
 - The Black–Scholes formula introduced a systematic way to value options using probability, volatility, and time — turning derivatives into a science.
- ◆ This innovation transformed financial markets, fueling the rise of modern derivatives trading and risk management.
- ◆ Just like algorithms power today's fintech platforms, Black–Scholes was a **mathematical breakthrough that unlocked new markets**





1998: WHEN THE MODEL BLEW UP

Long-Term Capital Management — the other half of the Black-Scholes story

THE SET-UP

A hedge fund founded in 1994.

On the board: Myron Scholes and Robert Merton — who receive the Nobel Prize in 1997, while the fund is running.

The strategy is the model itself, applied at very large scale and with very large leverage.

WHAT HAPPENED

1998. Russia defaults.

Markets stop behaving the way the model assumes they behave. Correlations that were independent all move together.

The fund loses around \$4.6 billion in four months and has to be rescued by a consortium of banks, coordinated by the Fed.

WHY IT IS HERE

Every innovation in this course creates a new risk that did not exist before it.

Black-Scholes did not fail because the maths was wrong. It failed because people forgot it was a model — a simplification, with assumptions.

Hold that thought until session 5, when we do the same exercise with AI.





CASE STUDY : THE RISE OF CREDIT CARDS

- ◆ Problem: Cash-based transactions were insecure and inefficient
- ◆ Solution: 1950 – Diners Club launches first credit card in NYC
- ◆ Result: Mass adoption and creation of an entire consumer credit infrastructure
- ◆ Impact: Payment systems, credit scoring, data-driven lending



CASE STUDY : HIGH-FREQUENCY TRADING

- ◆ Problem: Traditional stock trading was too slow to exploit tiny price differences between markets. Opportunities for arbitrage existed, but humans and legacy systems couldn't act fast enough.
- ◆ Solution: Deployment of high-speed algorithms and co-location (servers placed directly next to exchange servers) to execute trades in microseconds.
- ◆ Result: Firms captured arbitrage profits by buying low in one market and selling high in another almost instantly. Liquidity increased, and bid-ask spreads narrowed.
- ◆ Impact:
 - Redefined competition in trading: speed became the decisive edge.
 - Raised ethical and regulatory concerns over fairness and market manipulation.
 - Inspired global regulation (e.g., MiFID II in Europe, SEC scrutiny in the US).





CASE STUDY : BITCOIN

◆ Problem

- Traditional finance relies on intermediaries (banks, governments).
- 2008 crisis exposed fragility and mistrust in centralized systems.
- Digital payments existed but faced the *double-spending problem*.

◆ Solution

- Bitcoin whitepaper (Satoshi Nakamoto, 2008).
- Blockchain + Proof of Work: secure, decentralized transaction validation.
- Fixed supply (21 million coins) ensuring digital scarcity.

◆ Result

- First functioning peer-to-peer electronic cash system.
- Enabled global transfers without banks or central authority.
- Rapid adoption by early tech communities and investors.

◆ Impact

- Sparked cryptocurrency and blockchain revolution.
- Inspired thousands of fintech innovations and Central Bank Digital Currency (CBDC) projects.
- Ongoing debates on regulation, energy use, and role of money.

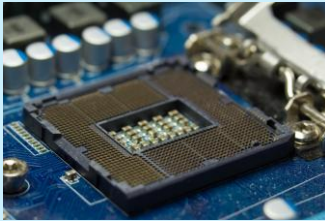


EARLY FINTECH

- ◆ Fintech is not a disruption : it's only the latest wave in a (very) long evolution
- ◆ What do all these innovations have in common?

WHY HAS FINANCE ALWAYS EVOLVED?

Tech Driven Innovation



- Telegraph and ticker tape (19th century) — speeding up information flow for traders.
- Blockchain to handle with international money transfer (Ripple)

Need Driven Innovation



- Ancient Mesopotamian clay tablets (c. 2000 BCE) — record-keeping to support loans and trade.
- Check writing in Renaissance Europe — need for secure and transferable payment methods.
- Buy Now – Pay Later to boost sales (Younited)

Regulation Driven Innovation



- Glass-Steagall Act (1933) — leading to new financial products outside commercial banking.
- Post-2008 financial crisis reforms — emergence of RegTech to comply with complex regulations.
- Digital KYB (DotFile)

◆ Key Point : the three often interact



2008: THE WINDOW THAT OPENED EVERYTHING

Why fintech started then — and not in 1999

2006

CLOUD

AWS launches.
Infrastructure cost for a
financial start-up falls
from millions to a credit card.

2007

DISTRIBUTION

The iPhone ships.
Suddenly every customer
carries a bank branch
in their pocket.

2008

TRUST

Lehman collapses.
The incumbents lose the one
thing they had that
start-ups could not buy.

2009+

RETREAT

Basel III forces banks to
hold more capital. They exit
SME lending and trade finance.
The field is left open.

None of the four would have been enough on its own.

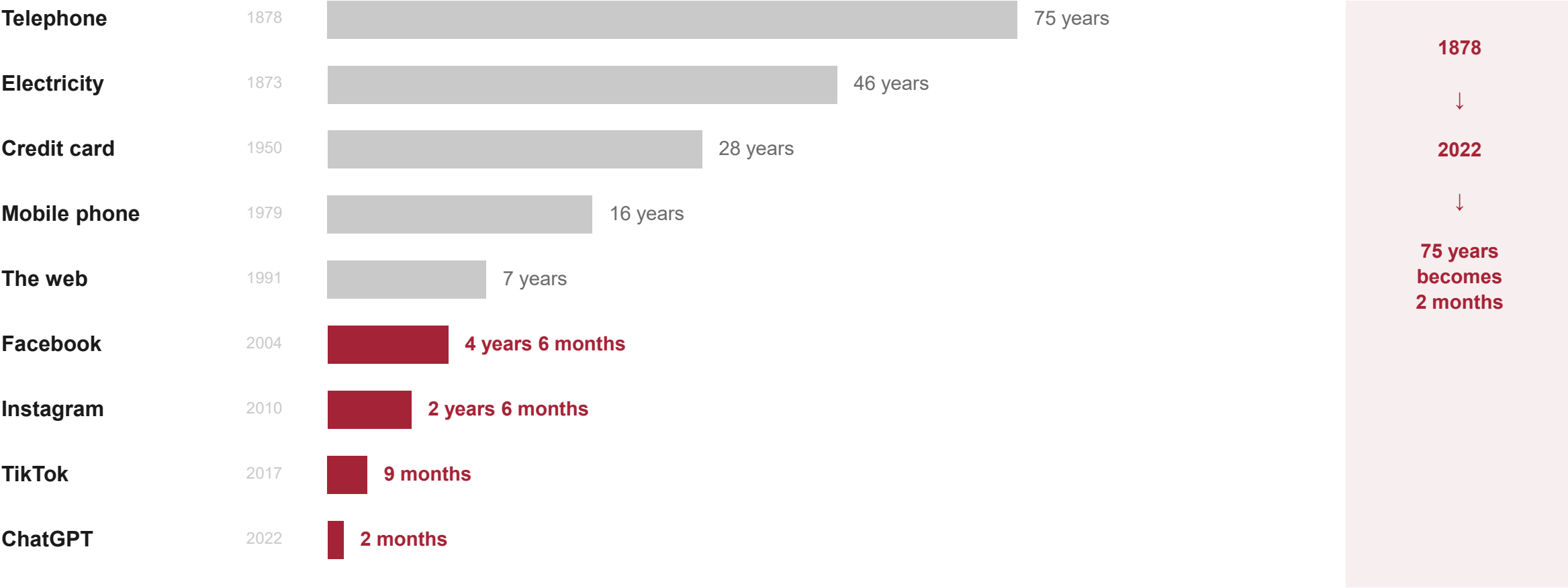
All four inside 36 months created an entire industry.





TIME TO 100 MILLION USERS

The same product idea, launched in different decades, spreads at wildly different speeds



Bars are scaled for readability, not linearly. Figures are widely-cited estimates — orders of magnitude, not audited numbers.





WHY IS IT ACCELERATING?

Four forces that compound — this is the mechanism behind every slide so far

1 The cost of trying collapsed

1995: launching a bank meant a licence, branches and a core banking system. Call it 100 million euros and five years.

2026: banking-as-a-service, white-label KYC, an API. Tens of thousands of euros and six months.

2 Finance became modular

Payments: Stripe. Accounts: Swan, Treezor. Identity: Onfido. Bank data: Powens, Plaid.

You no longer build a financial product. You assemble one. Every new brick multiplies the number of possible combinations — innovation turns combinatorial.

3 Regulation forced the doors open

PSD2 did not permit APIs. It made them compulsory.

The regulator imposed the modularity that made assembly possible. Keep this in mind every time someone tells you regulation only slows innovation down.

4 Distribution became instant

A Mesopotamian innovation travelled at the speed of a caravan. Double-entry bookkeeping took 200 years to cross Europe.

An app reaches 100 countries on day one, at no marginal cost.





IS FINTECH REALLY “NEW”?

- ◆ At the end of this historical part, we could ask one question :
 - What does “innovation” mean in finance:
 - ◆ new tech,
 - ◆ new thinking?



TRADITIONAL FINANCE

MAIN PAIN POINTS



Make an impact



ONBOARDING IN TRADITIONAL FINANCE

- ◆ **Lengthy paperwork** (IDs, proof of address, signatures)
- ◆ **In-person branch visits required**
- ◆ **Waiting days/weeks for account activation**
- ◆ **Poor UX:** multiple forms, redundant data entry



CROSS-BORDER PAYMENTS

- ◆ Slow (2–5 days via SWIFT),
- ◆ Expensive (high fees, FX spreads)
- ◆ Opaque (hard to track).
- ◆ Multiple middlemen (correspondent banks, clearinghouses) increase costs.



LOAN APPROVAL FOR A STARTUP

- ◆ Long approval process (weeks/months)
- ◆ Heavy collateral requirements
- ◆ Limited credit scoring for new businesses
- ◆ Bankers risk-averse → High rejection rate



TRANSPARENCY & TRUST

- ◆ Hidden fees
- ◆ Complex products
- ◆ Scandals (e.g., LIBOR manipulation,...).



SLOW INNOVATION & LEGACY SYSTEMS

- ◆ Old IT infrastructures (mainframes, COBOL code) make banks slow to innovate.
- ◆ Ten of thousands of employees, thousands of branches...



FRICTION EVERYWHERE: THE LIMITS OF TRADITIONAL FINANCE

- ◆ **High Costs :**
 - Legacy systems, fees, inefficiencies
- ◆ **Slow Processes :**
 - Manual KYC, long onboarding
- ◆ **Limited Access :**
 - Unbanked/underbanked, developing markets
- ◆ **Lack of Transparency :**
 - Hidden fees, product complexity
- ◆ **Customer Experience Gaps :**
 - Paperwork, delays, lack of personalization



1.3 BILLION ADULTS STILL HAVE NO ACCOUNT

And this is the one number in the whole course that is moving fast in the right direction

1.3bn

adults with no account
of any kind

79%

of adults now have one
— up from 74% in 2021

55%

of the unbanked
are women

650m

live in just
eight countries

WHY PEOPLE STAY OUTSIDE

- No ID or proof of address
- Distance — the nearest branch is hours away
- Cost of holding an account
- Digital literacy, and connectivity
- Gender: legal and social barriers to holding money

READ IT AS ACCELERATION

74% to 79% in three years. Hundreds of millions of people entering the formal financial system inside one presidential term.

Almost none of it came from banks opening branches. It came from mobile money — the M-Pesa model, repeated.

Source: World Bank, Global Findex Database 2025 (survey conducted 2024, 145,000 adults, 141 economies).





TWO CUSTOMERS THE SYSTEM WAS NOT BUILT FOR

Same root cause: the credit and distribution models assume a past you may not have

Small businesses

the €50,000 problem

THE PROBLEM

- Credit scoring reads historical accounts and collateral
- A three-year-old company has neither
- Manual underwriting, weeks of delay
- High rejection rates, so many stop asking

WHAT FINTECH DID ABOUT IT

- Alternative data: sales, invoices, card receipts, payment flows
- Dynamic scoring instead of a fixed grid
- A decision in minutes, inside the tool the business already uses

Small investors

the €500 problem

THE PROBLEM

- Products designed for wealthy and institutional clients
- High minimums, layered fees, intermediaries
- Little financial education, and jargon as a barrier
- Result: savings sit in low-yield accounts for decades

WHAT FINTECH DID ABOUT IT

- Fractional shares — €10 buys a slice of anything
- Robo-advisers: a managed portfolio at a fraction of the cost
- Mobile-first, and increasingly the first place people learn





FAST FRONT-END, SLOW PLUMBING

Fifteen years of fintech. Look honestly at what actually changed

WHAT GENUINELY BROKE OPEN

Opening an account	3 weeks → 8 minutes
Paying a friend	cheque → instant
A stock trade	\$10 → \$0
Reaching a customer	a branch → a phone
Sending €50 in the euro area	1 day → 10 seconds

WHAT DID NOT MOVE AT ALL

A cross-border payment	still 2 to 5 days
SWIFT, built in 1973	still routes the world
Core banking systems	still COBOL
The cheque, in France and the US	still legal tender
Bitcoin, 17 years old	still not a payment rail

The rule: the closer an innovation gets to the core of the system, the slower it moves.

Front-ends are rebuilt in months. Rails are replaced in decades. This explains most of the rest of this course.





WORKSHOP – IDENTIFY A FINANCIAL PAIN POINT

- ◆ You will form pairs and list 1 or 2 pain points you've personally faced

- ◆ **On this example answer the followings :**
 - Was it costly? Slow? Unfair? Confusing?
 - Could a digital solution have improved it?



TYPES OF FINTECH INNOVATION

Make an impact



ONE CAUSE, THREE DIMENSIONS

An innovation has a trigger, then deploys along three axes — do not confuse the two

WHAT TRIGGERS IT — PICK ONE

Technology

A new capability arrives and makes something possible that was not.

Ripple for cross-border settlement

Need

A pain point becomes unbearable, or a segment is left unserved.

BNPL to lift merchant conversion

Regulation

A rule changes and opens — or closes — a market.

PSD2 creating account aggregation

HOW IT DEPLOYS — ALL THREE, ALWAYS

Business model

How is it monetised, and by whom?

Klarna is free for you. The merchant pays.

Distribution

Where does the customer meet it?

Inside Shopify. Inside WeChat. Not in a branch.

Ethics & impact

Who benefits, who carries the risk?

Fossil-free banking. Or gamified trading.



TECHNOLOGY AT THE CORE

◆ Key Tech : Key Fintechs :

- **AI/ML:** Robo-advisors, fraud detection
- **Blockchain:** Smart contracts, tokenized assets
- **APIs:** Embedded finance, neobank-as-a-service
[Powens Customer Stories](#)
- **Mobile:** M-Pesa, Revolut

◆ Characteristics:

- Scalable
- Disruptive
- Needs strong engineering.

 **Wealthfront**

 **Betterment**

 **Chainalysis**

 **stripe**

 **PLAID**

 **Linxo**

 **pennylane**

 **LEDGER**

 **AGICAP**

Revolut

EXAMPLE : AGICAP X SWAN



INNOVATION FROM THE CUSTOMER'S PAIN

◆ Solve people's pain:

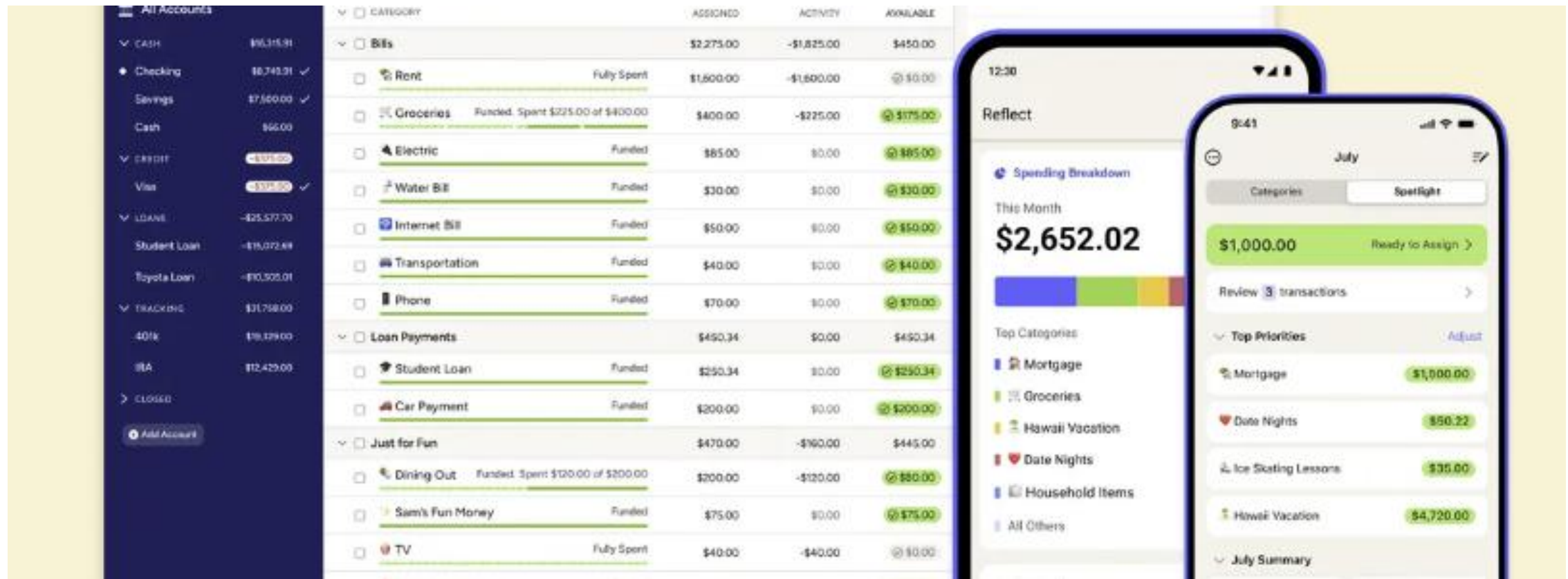
- **Credit Access:** Buy-now-pay-later
- **Financial Planning:** Budgeting apps
- **SMEs:** Invoice financing
- **Cross-border Transfers**

◆ Characteristics:

- Scalable
- Disruptive
- built with industry pain or user-centric vision.



EXAMPLE : YNAB YOU NEED A BUDGET



RULES THAT SPARK INNOVATION

◆ Regulations and what they enabled :

- **PSD2 & Open Banking (EU)** : Aggregators, neobanks, personal finance apps
- **Regulatory Sandboxes** : Fintech testing in the UK, Singapore, UAE
- **Crypto & DeFi Regulation** : New players in capital markets
- **Data Privacy Laws (GDPR, CCPA)** : Secure identity and consent solutions



◆ Characteristics:

- Key topic : Finance is regulated
- Compliance and legal shifts open space for new entrants.

Comply
Advantage

Conformitee

BUSINESS MODEL INNOVATION

- ◆ New ways of monetizing finance beyond traditional fees.
 - **Robinhood** → commission-free trading, revenue via Payment For Order Flow
 - **Klarna** → free consumer credit, monetized via merchant fees
 - Freemium apps like **Revolut** → basic free, premium subscription tiers

Robinhood 

Klarna.

Revolut

DISTRIBUTION DRIVEN INNOVATION

- ◆ Meeting customers where they are (not in bank branches)
 - Embedded finance in e-commerce → **Shopify Capital, Apple Pay**
 - Superapps (**WeChat, Grab**) → finance inside everyday apps



WECHAT : SUPERAPP INCLUDING PAIEMENTS





SOCIAL & ETHICAL INNOVATION

- ◆ Demand for responsible, sustainable finance.
 - Aspiration → fossil-free banking
 - ESG robo-advisors
 - Inclusive finance for women and migrants

GreenFi

 Patapia

Revolut



WHO'S ACTUALLY ON THE FIELD

Four groups, four business models — and only one of them is on the news

Incumbents

The banks

BNP Paribas · Société Générale
JPMorgan · Crédit Agricole

Buy, partner, or spin off a subsidiary (Nickel, Hello bank!).
JPMorgan spends around \$17bn a year on technology.
Goldman's Marcus shows it does not always work.

Challengers

The famous ones

Revolut · Qonto · N26
Lydia · Younited

Own the customer relationship, rebuild the experience.
Highest visibility, hardest economics — most are still fighting for profitability.

Enablers

The invisible ones

Stripe · Swan · Treezor
Onfido · Powens · Marqeta

Sell the bricks to everybody else. B2B, unglamorous, recurring revenue.
This is where the margins are — and where the jobs are.

Big Tech

The elephant

Apple Pay · Google Pay
Amazon Lending · Alipay

Own distribution and the device. No licence appetite, no legacy.
Meta's Libra was killed outright by regulators in 2022.

The column you never think about is the one that is hiring.



THE \$3 BILLION COUNTER-EXAMPLE

Goldman Sachs had everything an incumbent is supposed to need. It still lost.

WHAT GOLDMAN BROUGHT

- Unlimited capital
- No legacy retail systems to migrate
- World-class engineers
- A global brand
- Five years, and the board behind it

\$3bn+

lost by the consumer
division since 2020

then it sold
the whole thing

WHAT IT TURNED OUT TO NEED

Distribution

No retail customer relationship of its own. It had to rent Apple's — on Apple's terms.

Data

No history of consumer defaults. You cannot underwrite a borrower you have never seen.

An operating model

Built for a few thousand negotiated deals a year. Retail is millions of tiny ones, and the margin is the cost per account.

THE UNWIND

● 2016

Marcus launches

● 2019

Apple Card — Goldman carries all the credit risk

● 2021

GreenSky bought for \$2.2bn

● 2024

Marcus Invest, the GM card and GreenSky all sold

● 2025

Apple Card's \$20bn book goes to JPMorgan

If capital and technology were the constraint, this would have worked. So what is?



FRAMEWORK FOR UNDERSTANDING INNOVATION



Make an impact



SIX QUESTIONS FOR ANY FINANCIAL INNOVATION

Past, present or future — this is the grid you will use for the rest of the course

1

What is the pain point?

One sentence, one specific group.

Weak: "banking is complicated." Strong: "opaque FX fees make small cross-border payments uneconomic."

2

Who is the user?

A segment and a constraint they live with.

Weak: "everyone." Strong: "freelancers invoicing clients in another currency."

3

What is the solution?

The value proposition, in the user's words.

If you cannot say it without jargon, you have not understood it.

4

What technology carries it?

Name the load-bearing part.

Often less exotic than the pitch deck claims. Sometimes there is none.

5

What is the regulatory position?

Licence, supervisor, exposure.

In finance, "none" is never the answer. It is a risk they have not priced.

6

What changed, at what speed?

Adoption, and the effect on everyone else.

The question that separates a product from an innovation.

If you cannot answer question 1 in one sentence, nothing after it will be worth much.



CASE STUDY: M-PESA (KENYA)

M-Pesa is a mobile phone-based money transfer, payment, and micro-financing service launched in Kenya in 2007 by Vodafone. It allows users to deposit, withdraw, send money, and pay for goods and services using simple SMS technology—no bank account or internet connection required. M-Pesa quickly became a vital tool for financial inclusion, especially in rural and underserved communities, and has since expanded to multiple countries. By transforming mobile phones into wallets, it revolutionized how millions of people access financial services.



- ◆ **Pain Point:** Lack of banking infrastructure in rural areas
- ◆ **Target User:** Unbanked and underbanked populations in Kenya
- ◆ **Solution:** Mobile phone-based money transfer and payment system
- ◆ **Technology:** SMS-based mobile payment platform
- ◆ **Regulatory Context:** Backed by the Kenyan Central Bank, light regulation initially
- ◆ **Adoption Curve:** Fast adoption → >70% of adult population uses it



YOUR TURN: RUN THE FRAMEWORK

Six questions, two companies, six minutes — then you present, not me

THE BRIEF

1 Pairs.

Odd rows take Revolut. Even rows take Robinhood.

2 6 minutes.

Answer the six questions of the framework. Written, in note form.

3 Then 60 seconds.

One pair per company presents. The rest of the room challenges.

4 The one that counts.

Question 6 — adoption and impact. Anyone can describe a product. Tell me what it changed for everybody else.

What a weak answer looks like

"It's an app for banking." — that is a category, not a pain point.

"Everyone." — that is not a target user.

"Blockchain." — name the technology that is actually load-bearing.

No regulatory answer at all — in finance that is never the right answer.

Vague answers get challenged. Fairly.





CASE STUDY: REVOLUT (UK)

Revolut is a UK-based neobank launched in 2015 that offers global banking services through a mobile app. It provides multi-currency accounts, real-time currency exchange, crypto trading, budgeting tools, and stock investments—all in one platform. Revolut targets digital-savvy consumers and travelers, emphasizing low fees, transparency, and user control. By leveraging APIs, real-time analytics, and a mobile-first experience, Revolut challenges traditional banks with a fast, flexible, and borderless financial ecosystem.



- ◆ **Pain Point:** Expensive FX fees and lack of financial flexibility abroad
- ◆ **Target User:** Young professionals, travelers, digital natives
- ◆ **Solution:** Borderless bank account with free FX, spending control, crypto features
- ◆ **Technology:** API-based banking, mobile-first UX
- ◆ **Regulatory Context:** Started with e-money license, then banking license in EU
- ◆ **Adoption Curve:** >60 million users globally, “super-app” ambition

CASE STUDY: ROBINHOOD (US)

Robinhood was launched in the US in 2013 with a bold promise: to democratize finance by eliminating brokerage fees and making investing accessible to everyone. Through its mobile-first, easy-to-use platform, it attracted millions of young retail investors, many of them first-time traders. Robinhood's rise reflects both the disruptive power of fintech and the challenges that come with innovation, from gamification risks to heightened regulatory scrutiny.



- ◆ **Pain Point:** Retail investors faced high brokerage fees and barriers to entry in stock trading.
- ◆ **Target User:** Young retail investors, beginners in the US.
- ◆ **Solution:** Commission-free stock & crypto trading app with simplified UX.
- ◆ **Technology:** Mobile app, API-driven order execution, gamified interface.
- ◆ **Regulatory Context:** SEC & FINRA oversight; scrutiny after GameStop saga for payment-for-order-flow practices.
- ◆ **Adoption Curve:** Explosive growth from 2013 to 2021, then increased regulatory oversight and user churn.



CASE STUDY: QONTO (FRANCE)

Qonto is a French neobank founded in 2016 with the mission to simplify business banking for entrepreneurs, freelancers, and SMEs. Traditional banks often overlooked this segment, offering clunky tools and slow processes. By leveraging digital platforms and open banking, Qonto built a user-friendly solution for expense management, invoicing, and team collaboration, quickly positioning itself as a leader in European business banking.



- ◆ **Pain Point:** SMEs and freelancers faced poor banking UX, slow processes, and limited tools for expense & team management.
- ◆ **Target User:** Startups, SMEs, freelancers in France.
- ◆ **Solution:** Digital-first business banking with invoicing, expense management, cards, and multi-user access.
- ◆ **Technology:** Mobile-first UX, APIs, open banking integration.
- ◆ **Regulatory Context:** PSD2-compliant, French e-money license, operates under EU regulatory frameworks.
- ◆ **Adoption Curve:** Rapid growth since 2016 → +400k clients, strong adoption among entrepreneurs.



A DEALING ROOM: 1996 VS 2026

I have worked in both. This is the acceleration you have been looking at, from the inside

1996

On the desk

Two phone handsets, a Reuters terminal, a calculator, a paper pad.

A price

You shouted. Someone shouted back.

An option quote

Spreadsheet, or by hand.
Minutes, sometimes hours.

The record

A paper ticket in a box.
Confirmation by fax the next morning.

The room

Loud enough that you lost your voice
by Wednesday.

2026

On the desk

Six screens. Nobody is holding a phone.

A price

A machine quotes, a machine answers.
Microseconds.

An option quote

Priced continuously, in real time,
across the whole book.

The record

Timestamped, immutable, recorded,
readable by the regulator.

The room

Quiet. Genuinely quiet.

What disappeared: the noise. What appeared: latency. Thirty years — one working life.





WHAT'S COMING DOWN THE PIPE

Not predictions — trajectories that are already funded and already being built

Agentic payments

An AI agent opens the account, compares the products, and executes. It has no brand loyalty, does not see your advertising, and reads every line of the terms.

If your customer is a machine, what is a bank's brand worth?

Stablecoins in corporate treasury

Not retail speculation — treasurers settling supplier invoices in minutes instead of days, under MiCA in Europe.

Does this finally fix the cross-border problem, or just move it?

Tokenised money market funds

BlackRock and others issuing fund units on-chain. Collateral that moves in seconds instead of overnight.

The plumbing changing — which is why it will be slow and why it matters.

The digital euro

A central bank issuing directly to the public would put the ECB in a relationship it has never had with citizens.

Who loses deposits if this ships?

Check the status of each of these the week before you sit an exam on them — three of the four are moving targets.





WHAT WILL LOOK LIKE A CLAY TABLET IN 2050?

Closing workshop — five minutes

**In pairs. Name three things about finance in 2026
that will look as primitive in 2050
as a clay tablet looks to you today.**

One rule: at least one of the three has to be something you personally used this week.

HOW IT RUNS

4 min in pairs, three written proposals

4 min each pair gives its strongest one

1 min the room votes for the most likely

Kept we reopen this list in Lecture 8

THE POINT OF THE EXERCISE

For two hours you have looked at this movement from the outside, as history. You are standing inside it. Somebody in 2050 will put your banking app on a slide next to a clay tablet.





WRAP-UP



Make an impact



WHAT TO TAKE OUT OF TODAY

Five things — the fourth is the one that will make you sound like an analyst

1

Finance has always been technological.

Clay tablets, double-entry, the telegraph, Black-Scholes, blockchain. Fintech is not a break with the history of finance — it is the current position of a very long line.

2

But the line is getting steeper.

The gap between major innovations has gone from millennia, to decades, to months. That part genuinely is new.

3

Four forces explain it.

The cost of trying collapsed. Finance became modular. Regulation forced the doors open. Distribution became instant.

4

The front-end is fast. The plumbing is not.

Cross-border payments still take days. SWIFT is 53 years old. Anyone selling you a revolution has not looked at the rails.

5

Every innovation creates a new risk.

Black-Scholes gave us LTCM. Free trading gave us gamification. Ask what a new tool makes possible that was not possible before — in both directions.

Before next week: pick one fintech you actually use. Run the six questions on it. One page.





PAST, PRESENT & FUTURE

Which of the following best illustrates that innovation has always been a core part of financial systems?

- A. The creation of blockchain in 2008
- B. The invention of credit cards in the 1950s
- C. The use of clay tablets for accounting in Mesopotamia
- D. The rise of robo-advisors after 2010



PAST, PRESENT & FUTURE

Which of the following is not typically cited as a pain point of traditional financial systems?

- A. High cost of access
- B. Over-personalization
- C. Lack of financial inclusion
- D. Inefficient processes



PAST, PRESENT & FUTURE

What is the main characteristic of embedded finance?

- A. Offering financial services through mobile-only banks
- B. Integrating financial services into non-financial apps or platforms
- C. Using blockchain for identity verification
- D. Replacing central banks in money issuance



PAST, PRESENT & FUTURE

A start-up rebuilds account opening from three weeks down to eight minutes, but the transfer it launches still takes three days to reach a supplier abroad. Why?

- A.** The start-up did not invest enough in technology.
- B.** Cross-border payment is a coordination problem across many institutions, not a design problem inside one.
- C.** European regulation forbids faster international transfers.
- D.** Blockchain has already solved this and the company chose not to use it.





PAST, PRESENT & FUTURE

Which of the following fintechs is best known for solving the pain point of high FX fees for international transfers?

- A. Stripe
- B. Klarna
- C. M-Pesa
- D. Wise (formerly TransferWise)



Make an impact

